LLM Context Compression

A Knapsack Optimization Problem

Proposal Presentation

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Advanced Analysis of Algorithms (CS 8045)
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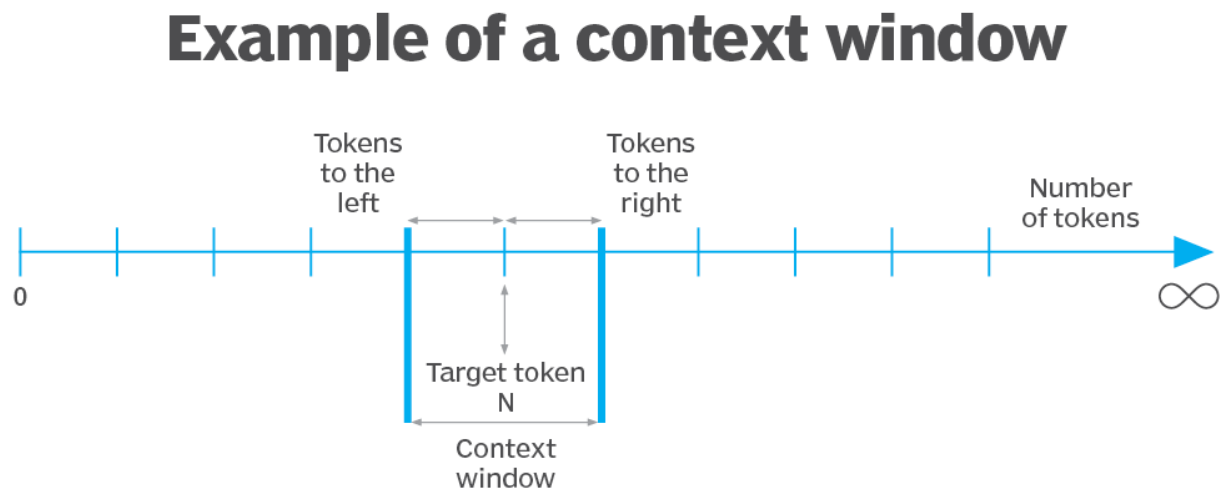
Presentation Overview

- Problem Statement: Context Window Limitations
- Mathematical Formulation: The 0/1 Knapsack Approach
- Proposed Algorithms: Exact DP vs. Heuristics
- Simulation Settings & Programming Environment
- Evaluation Metrics & Analysis Pipeline
- Project Milestones & Timeline

The Context Window Challenge

Large Language Models (LLMs) have a **fixed context window size**, typically ranging from 4k to 128k tokens.

- Long documents exceed these limits.
- Requires truncation or summarization.
- Goal: Retain maximum relevant info.



Mathematical Problem Statement

B

Token Budget

Given $\boldsymbol{n}$ chunks of text, each with size $\boldsymbol{s_i}$ and utility $\boldsymbol{u_i}$:

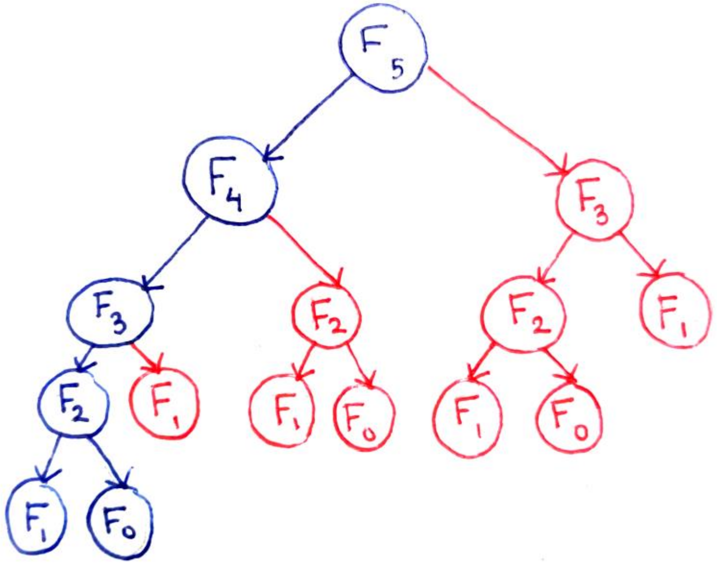
$$\text{Maximize } \sum u_i \text{ subject to } \sum s_i \leq B$$

This is functionally identical to the **0/1 Knapsack Problem**, which is NP-hard.

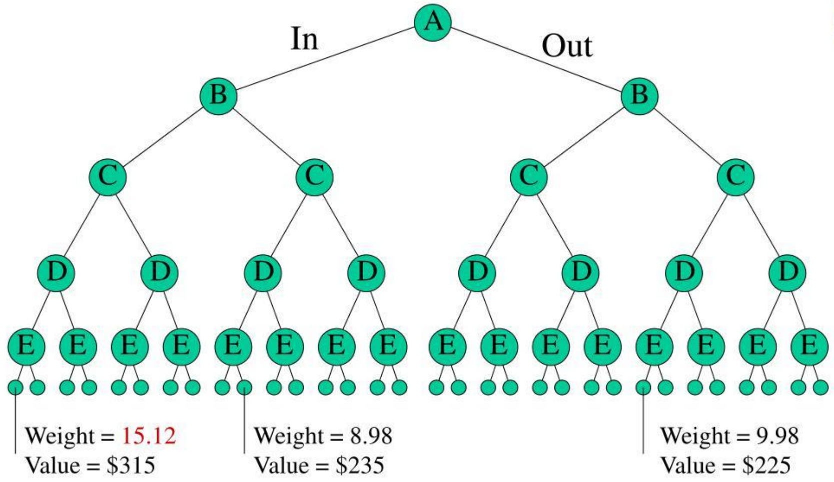
Algorithm 1: Exact Dynamic Programming

A classic pseudo-polynomial approach used as our **baseline for optimality**.

- Complexity: $O(n \times B)$
- Guarantees optimal subset.
- Efficient for current LLM limits.



Brute Force: Branching



Algorithms 2 and 3
Heuristics for Scale and Speed

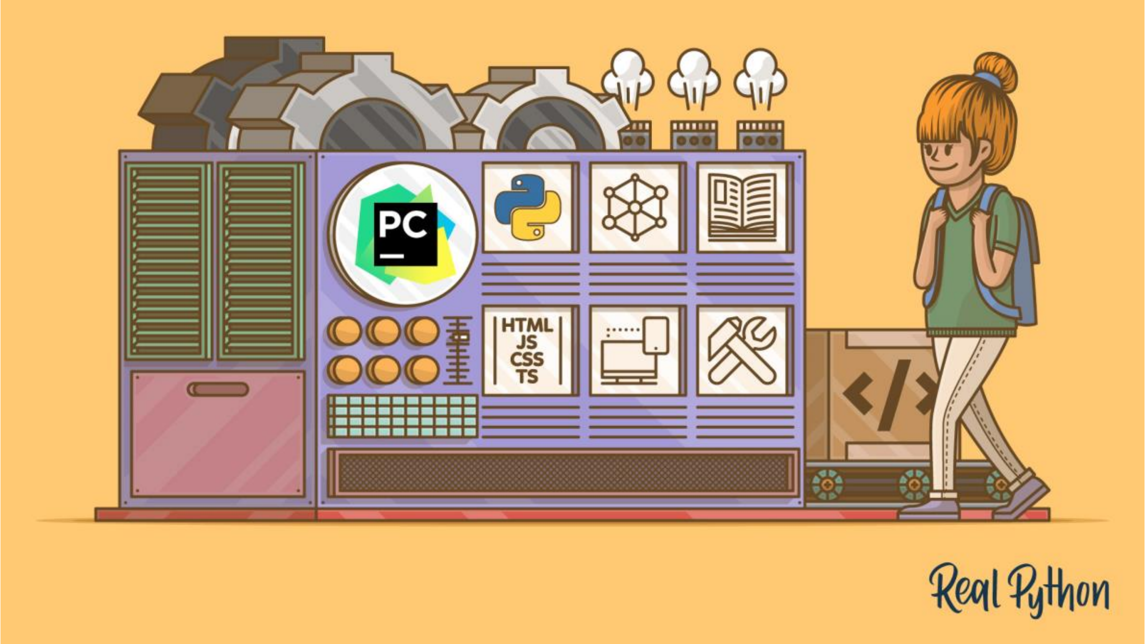
Greedy by Utility Ratio

Sorts by $\boldsymbol{u_i}$ / $\boldsymbol{s_i}$. Complexity: $O(n \log n)$. Fast heuristic.

Greedy with Refinement

Applies **Local Search** (swaps/adds/removes) to the greedy solution. $O(n^2)$ to improve quality over plain greedy.

Experimental Simulation Setup



Programming Environment & Setup:

- Python 3.10+ (Prototype & Evaluation)
    - Data Processing: Hugging Face, NumPy, Pandas
    - Text Processing: spaCy sentencizer or NLTK
- Utility score: embedding models, TD-IDF
- Budget B: 2k, 4k, 8k

Utility Computation & Data Sources

Utility Scoring

Calculated using a hybrid of **Sentence Embeddings** (semantic similarity) and **TF-IDF** (keyword matching).

Datasets

- **LongBench:** Long-context QA
- **HotpotQA:** Multi-hop reasoning
- **WikiText-2:** Synthetic long texts

Key Evaluation Metrics

- Total Utility Achieved
- Optimality Gap
- Token Budget Utilization
- Compression Ratio
- Runtime
- Memory Usage
- Empirical Time-Complexity Scaling

Project Milestones & Timeline

Week 5-6
Formalization & Data Pipeline

Week 7
Exact DP

Week 10-11
Greedy
Greedy Refinement

Week 12-13
Simulation & Visualization

Week 14
Final Report & Presentation

Project Team & Responsibilities

Amir

Overall coordination, communication, and implementation of the **Exact Dynamic Programming** algorithm.

Jay

Documentation, presentation management, and implementation of the **Greedy by Utility Ratio** heuristic.

Reuven

Codebase management, simulation results, and implementation of **Greedy with Refinement** (Local Search).

Questions?